# Tsz Fei Wang

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## Education

**University of Waterloo** | Honours Computer Science (Co-op) — Sept 2026 – May 2031 (expected)

## Skills & Technologies

**Languages & Frameworks**: C++, C, Python, Java
**Tools**: STM32 (HAL), SPI, PWM, ADC, KiCad, Freerouting, Arduino, Linux, Git, Docker, GitHub Actions

## Experience

**Waterloo Aerial Robotics Group (WARG)** — Sept 2026 – Present
*EFS Team Member* — *Waterloo, ON*

- Built the status LED indicator for the M10 GPS module on the team's aircraft, so GPS status can be read at a glance before flight.
- Programmed an STM32 Nucleo board in **C++** to read a potentiometer through an MCP3004 ADC over SPI and convert it into a 50 Hz servo PWM signal on TIM1, then refined the timer and SPI handling through code review.

**Century Frontier Asset Management** — Jul 2026 – Aug 2026
*Quantitative Research Intern* — *Shenzhen, CN*

- Designed a point-in-time backtesting pipeline in **Python** with **pandas** and yfinance that only shows each model data from before its prediction date, removing lookahead bias from LLM-driven equity research.
- Built a model-agnostic evaluation framework benchmarking **9 LLM architectures** across **104 tickers**, scoring signal quality separately from stated confidence and showing that confidence was a poor guide to accuracy.

**Krembil Research Institute, World Community Grid** — Mar 2026 – Jun 2026
*Software Engineer Intern* — *Toronto, ON*

- Built the full-stack data access portal for Catrin, serving **~4 million** TF-TG associations across **33,000 genes** to researchers.
- Reconciled the schemas of **9 independent data sources** so the portal can query them as one dataset.

**Mackenzie Cybersecurity Team** — Jun 2025 – Jun 2026
*Team Captain, President* — *Toronto, ON*

- Captained the team to **First in Canada** in CyberTitan, Canada's largest cyberdefense competition, securing Windows, Linux and Cisco systems.

## Projects

**Technova Hackathon Badge** | *KiCad, Hardware Lead* — *Oct 2026*

- Designed the attendee badge PCB for a **~300-person** hackathon in **KiCad**, integrating a RISC-V MCU, NFC tap-to-phone check-in, an OLED display, addressable LEDs and USB-C LiPo charging.
- Routed the 2-layer board (**1,300+ tracks**) and prepared the JLCPCB assembly files, keeping cost to about **$7 per badge** with in-stock LCSC parts.

**Submerge** | *GLSL, QML, Python* — *Oct 2026 – Present*

- Developed an animated koi theme for KDE Plasma 5 and 6 in **GLSL** and **QML**, with a shader water simulation and a live system HUD, reaching **~1,290 downloads** on the KDE Store.
- Added power-aware frame pacing for battery life and fixed blank rendering on NVIDIA/Wayland by porting the shaders to GLSL ES.

**Physical Studio** | *Python, React, Three.js, Docker* | ***Winner**, Hack the Valley 11* — *Oct 2026*

- Engineered the **Python** MCP server that lets an LLM design hardware projects from a library of measured parts and validate its own designs for collisions and clearance.
- Built the **React Three Fiber** viewer that animates each assembly step and deployed it with **Docker Compose** on Vultr.

## Awards & Honors

**Gold & Silver Medalist**, IT Systems Networking, Skills Ontario (Regional & Provincial) — Mar 2026 – May 2026
**National Champion x2**, 1st in Canada, Cybersecurity, $1,000 scholarship — May 2025, Mar 2026
**National Finalist**, 2nd in Canada, Cybersecurity, $1,600 scholarship — Mar 2025